

Definition 0.1 (Mutual Information): Let $P_{X,Y}$ be a joint probability measure on $(\mathcal{X} \times \mathcal{Y}, \mathcal{F} \otimes \mathcal{G})$ with marginals P_X and P_Y . The Mutual Information $I(X;Y)$ is defined as

$$I(X;Y) = D_{\text{KL}}(P_{X,Y} \| P_X \times P_Y).$$

If $P_{X,Y} \ll P_X \times P_Y$, then in terms of densities relative to a product reference measure $\mu \times \nu$,

$$I(X;Y) = \int_{\mathcal{X}} \int_{\mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} d\mu(x) d\nu(y).$$

Remark 0.2: Mutual information is symmetric, $I(X;Y) = I(Y;X)$, and non-negative. $I(X;Y) = 0$ if and only if $P_{X,Y} = P_X \times P_Y$, meaning X and Y are independent.

Remark 0.3: Unlike Shannon entropy or joint entropy, mutual information is a truly intrinsic functional of the joint measure $P_{X,Y}$. Mutual information is invariant under measure-preserving isomorphisms of the underlying probability spaces, and its value does not depend on the choice of reference measures μ and ν .

Remark 0.4: Mutual information can be expressed through the inclusion-exclusion-like relations:

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = H(X) - H(X|Y) = H(Y) - H(Y|X).$$

These identities hold provided the constituent entropies are finite.

Remark 0.5: In the context of f -divergences, mutual information is the specific case where $f(t) = t \log t$. Other "generalized mutual informations" can be defined by taking $D_f(P_{X,Y} \| P_X \times P_Y)$ for different convex functions f .

0.0.1 See also

0.0.2 References